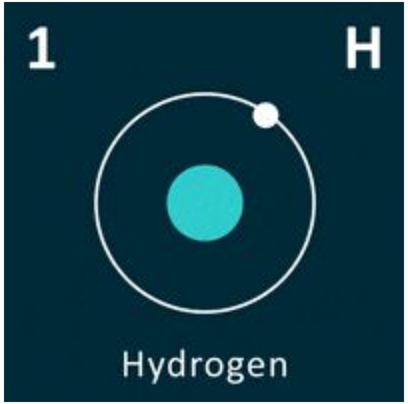
Neutralisation Circles

Neutralisation occurs when:

- A substance with a **low pH (acid)** reacts with a **base (high pH)** to form a neutral solution **(middle pH)**.

But what makes a substance alkaline or acidic?



Hydrogen without electron.

H^+

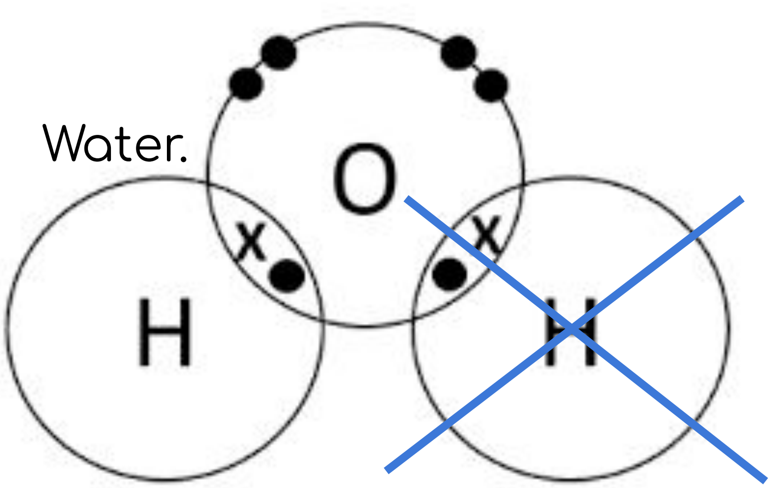
Hydrogen ion.

It is the number of H^+ that determines how acidic a substance is.

Acids have **MORE!** hydrogen ions.

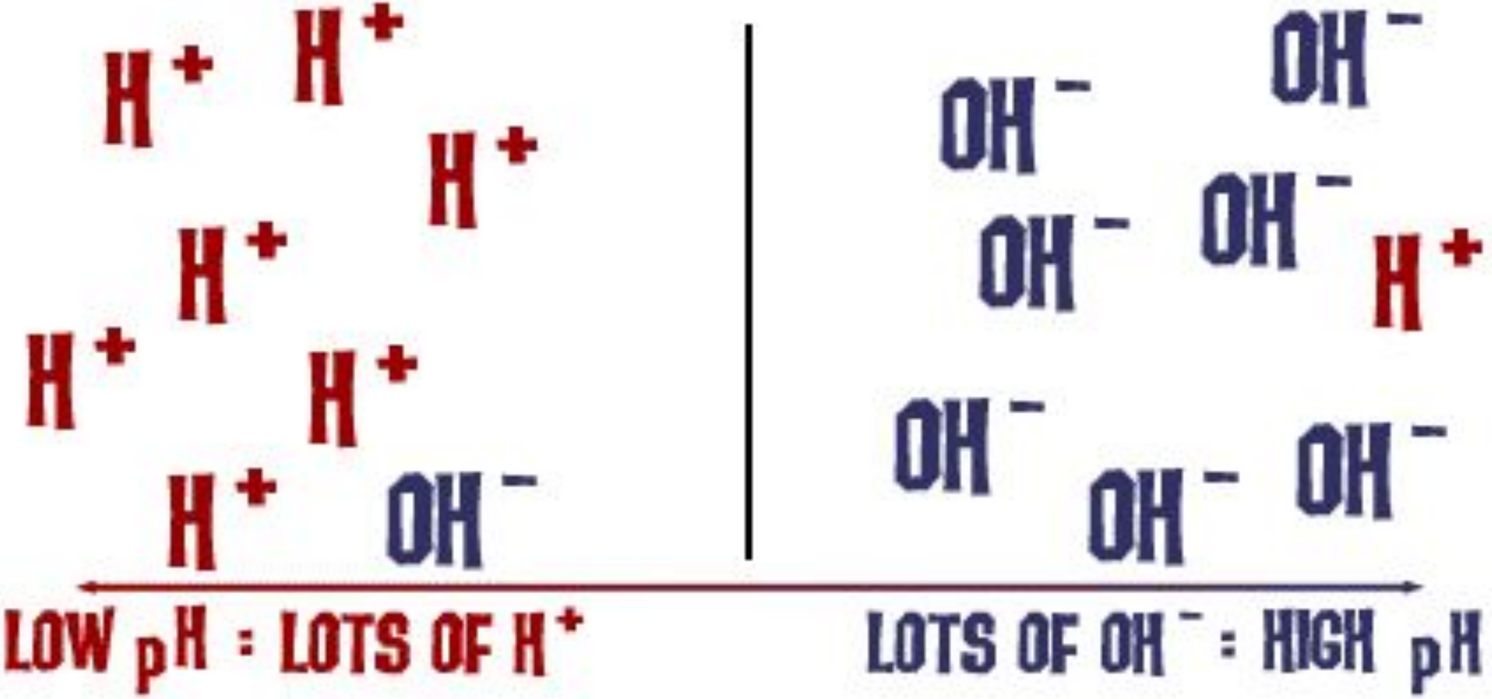
But what makes a substance alkaline or acidic?

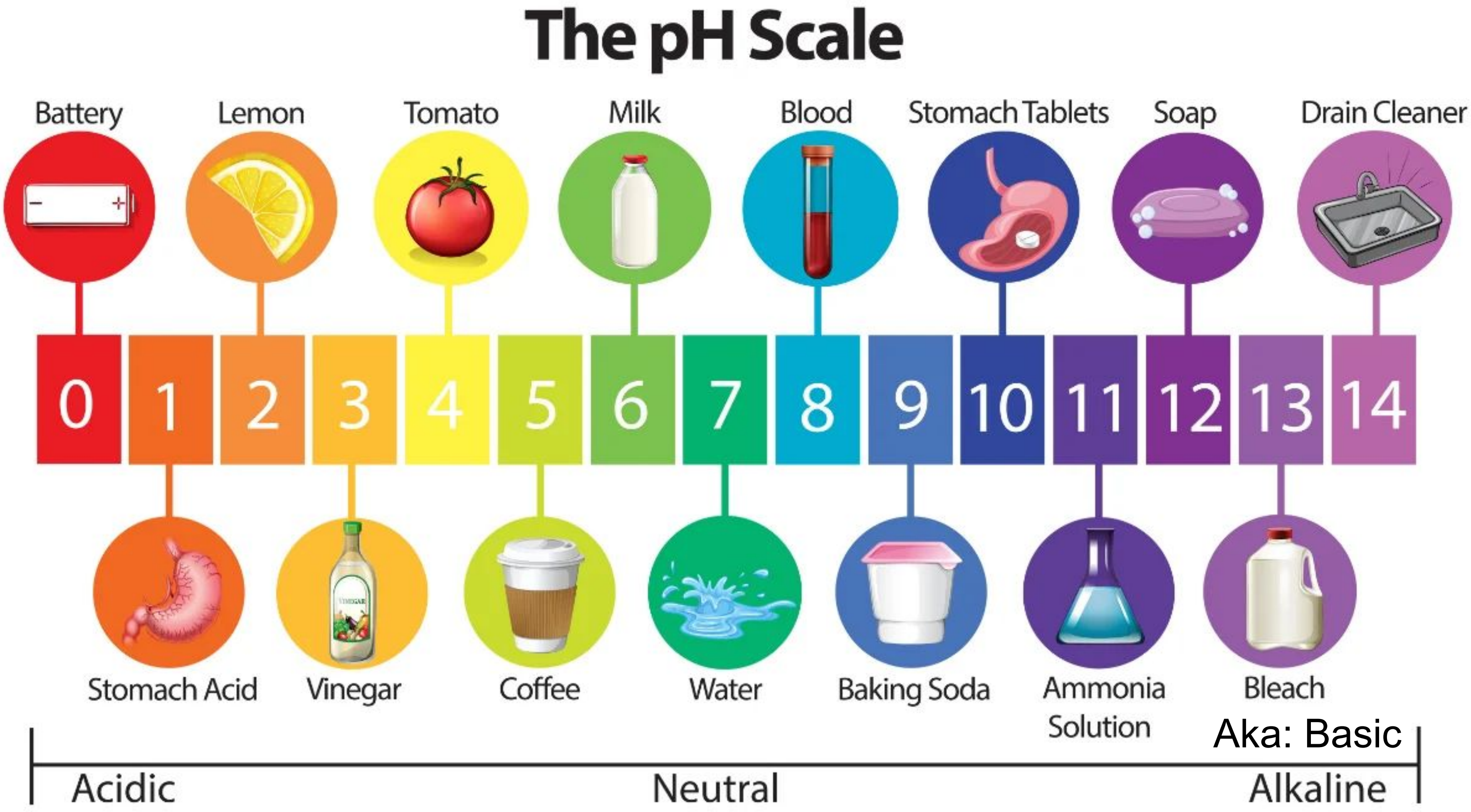
Water.



It is the number of OH^- that determines how alkaline a substance is.

Acids have **MORE!** hydroxide ions.


LOW pH : LOTS OF H^+
LOTS OF OH^- : HIGH pH


The pH Scale
Battery
Lemon
Tomato
Milk
Blood
Stomach Tablets
Soap
Drain Cleaner
0
1
2
3
4
5
6
7
8
9
10
11
12
13
14
Stomach Acid
Vinegar
Coffee
Water
Baking Soda
Ammonia Solution
Bleach
Acidic
Neutral
Alkaline

Aka: Basic


OH^-
H^+
H_2O

The Experiment

Equipment

- Eye protection
- Three dropping pipettes
- A pencil
- A white tile
- A sheet of filter paper, approximately 12.5 cm diameter
- Hydrochloric acid 0.1 mol dm–3
- Sodium hydroxide, 0.1 mol dm–3
- Universal indicator solution

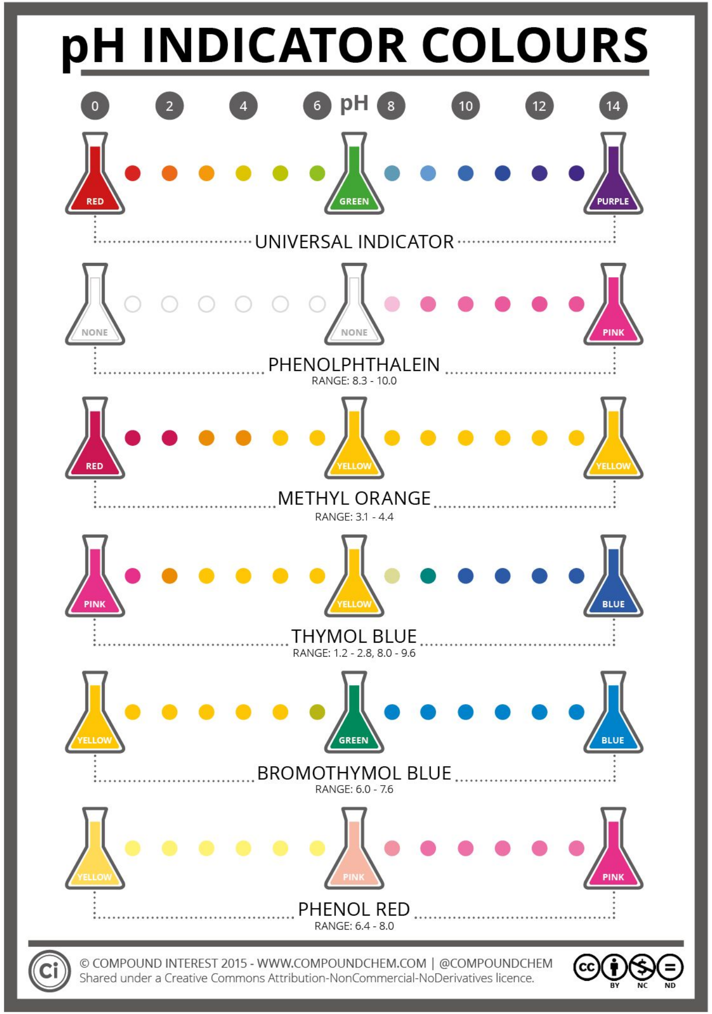
Method

1. Get filter paper.
2. Draw two pencil circles and label, ‘acid’ and ‘alkali.’
3. Place paper on white tile.
4. Use dropping pipette to dispense substances onto paper.
    a. The amount of each solution should be about same.
5. Solution spreads out.

Feel free to make nice patterns.

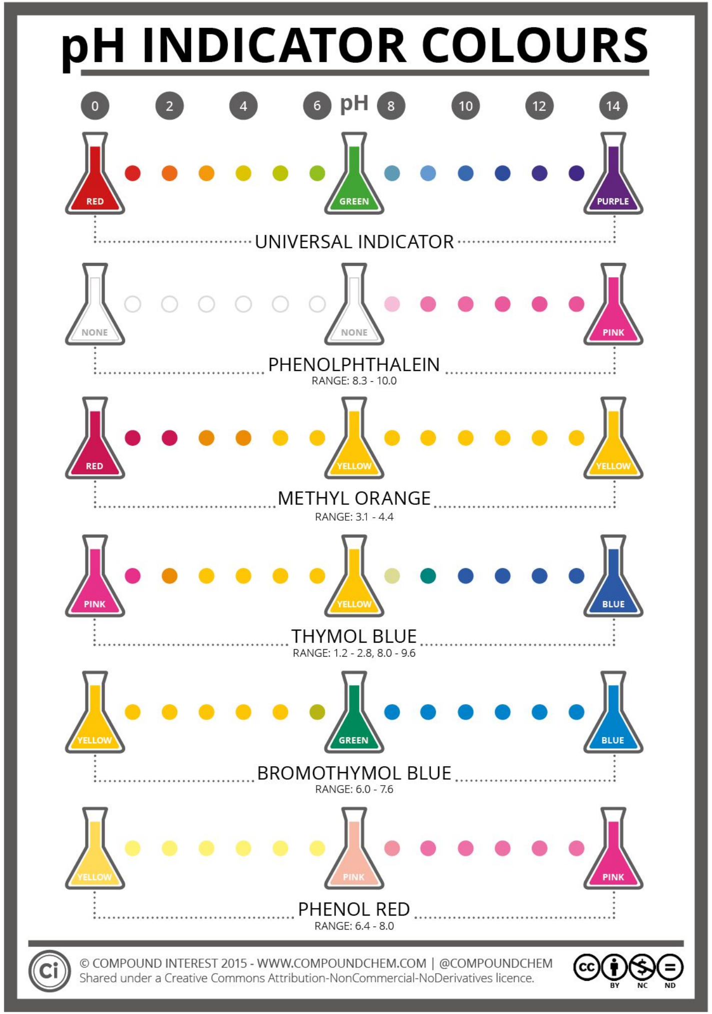
But how does the universal indicator know?

- Universal indicator is a **mix** of all the different indicators.
- These indicators change colours at different pHs.



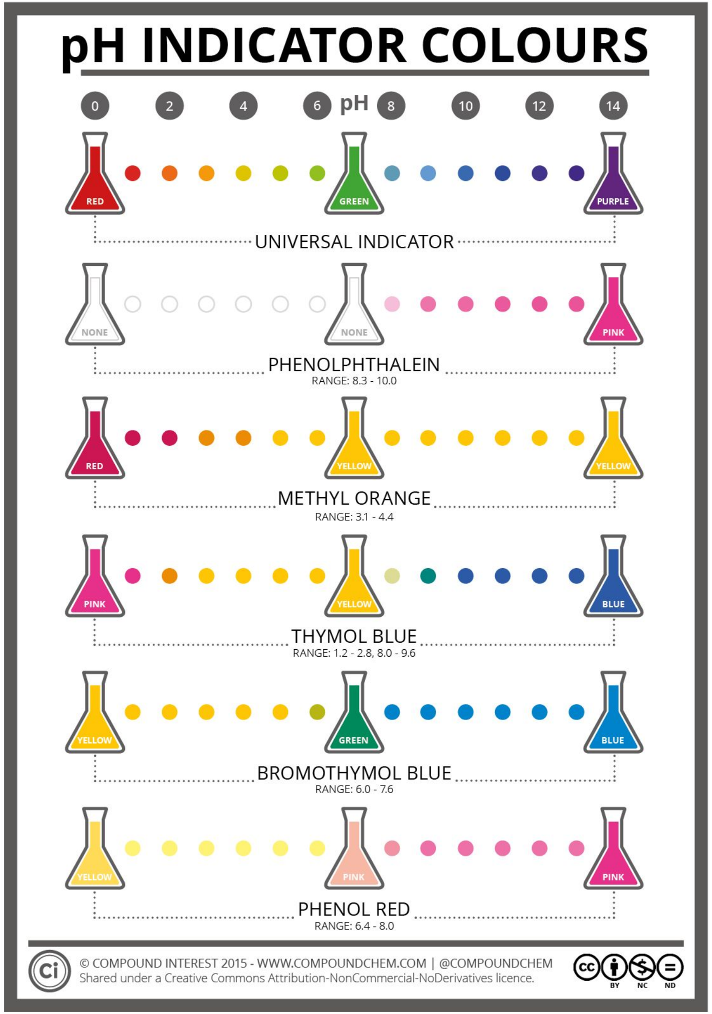
But how does the universal indicator know?

- The indicator molecule is added to the solution.
- The molecule’s structure changes.



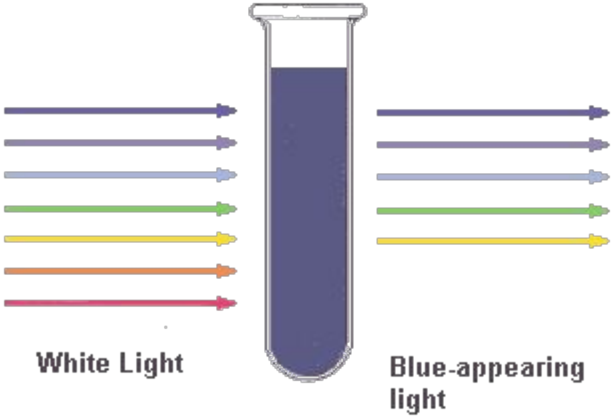
But how does the universal indicator know?

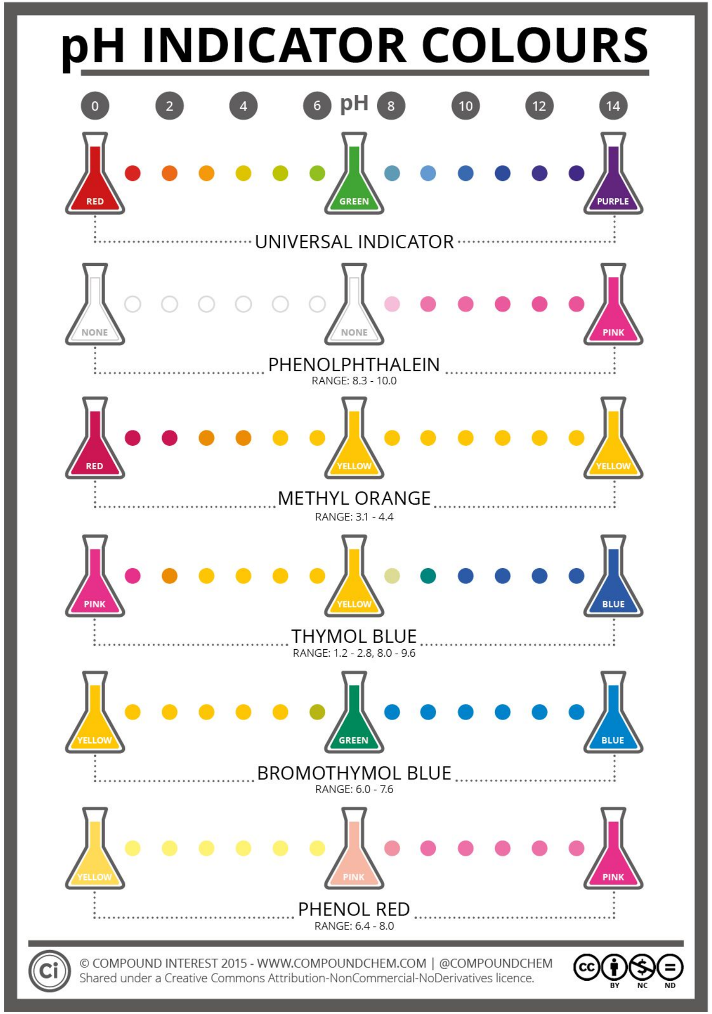
- This change depends on the number of:
    - OH^- ions.
    - H^+ ions.



But how does the universal indicator know?

- This changes the way it absorbs light.
- Changing its colour.





THANK YOU FOR COMING, me and Maria really appreciate!